

Mahalakshmi Ashokkumar

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Professional Summary

8 years of experience in the Automotive Embedded Systems domain with deep expertise in AUTOSAR BSW development, migration, and debugging for ECUs and VCUs across Powertrain and EV projects. Currently leading BSW migration on ST5 controller platforms as the SPOC for multiple OEM programs. Recognized as a specialist in advanced BSW debugging (stack overflow/underflow resets, boot control, CAN anomalies). Proven track record of leading teams, delivering high-quality software for global OEMs (Renault, PSA, RSA, Mahindra, Skoda, Hyundai), and collaborating onsite with Bosch France.

Core Competencies

- AUTOSAR BSW Modules: ComStack, IOStack, MemStack, BootCtrl, MCAL integration
- Controllers & Platforms: ST5 SR6P6, Infineon Tricore (40nm), RTA-OS, Linux
- Communication & Debugging: CAN, LIN, SPI, XCP, XETK, Lauterbach (JTAG), GDB, Serial Logs
- Tools & Frameworks: CANoe, Canalyzer, ETAS INCA/Ehooks, Ecu.Worx, SDOM, MATLAB, LabView
- Programming & Quality: Embedded C, C++, Python, MISRA compliance, Static Analysis
- Development Practices: Agile/Scrum, Requirements Analysis (DOORS, RQOne), Jenkins CI/CD, Git

Professional Experience

Robert Bosch Engineering, Bangalore – Lead BSW Engineer

Aug 2020 – Present

- SPOC for BSW migration across 3 projects, ensuring seamless transition to ST5 controller platforms.
- Delivered end-to-end AUTOSAR BSW configurations (ComStack, IOStack, MemStack, BootCtrl).
- Debugging Specialist – solved critical reset and stack overflow issues, designed reusable debugging workflows, reducing issue resolution by ~30%.

- Led a 3-member team, mentoring juniors in BSW concepts and ensuring delivery quality.
- Coordinated with customers & cross-site teams, driving requirement closure and project alignment.
- Onsite (France, 4 months): Supported STLA project, collaborating with Bosch France on ECU/VCU BSW development.

Key Projects:

- Renault (2023–Present): BSW Coordinator – Led migration & integration of BSW modules on ST5, SPOC for global clarifications.
- Gasoline ECU (2022–2023): BSW Responsible – Developed and configured ComStack, MemStack, IOStack; integrated MISRA-compliant C code.
- EVCU – PSA (2021–2022): BSW Developer – Implemented customer flashing over CAN, configured CAN transceivers (TJA1044/45, CY329), supported XCP/XETK profiling and prototyping with ETAS Ehooks.

Visteon Technical & Services Centre, Chennai – Software Engineer

Sep 2017 – Jul 2020

- Designed and developed HMI features in C/C++ on Linux for infotainment and cluster systems (PSA, Skoda, Mahindra, Hyundai).
- Integrated diagnostics, Bluetooth, media, display, Android Auto & Apple CarPlay.
- Awarded Spot Award (CES 2020) for rapid development of entry-level infotainment prototype in 2 weeks.
- Optimized SQLite-based phonebook module and diagnostic stack, improving system responsiveness.
- Worked with Wayland Compositor, X-server, EGL/OpenGL for rendering and cluster graphics.

Education

B.E. – Electronics & Communication Engineering, Anna University, 2017 (CGPA 7.71)

Awards & Recognition

- Rising Star Award – Bosch (2024): Honored as the Best Contributor in BSW topics, recognized for driving migration activities and solving complex debugging challenges.
- Spot Award – Visteon (2020): Delivered infotainment prototype showcased at CES 2020.

- Recognized as BSW Debugging Specialist within Bosch Global Software Technologies.

Additional Information

Languages: English, Tamil

Willing to relocate globally for automotive projects (Germany/Europe/US)